

PSYCH 390: Electrophysiology of Complex Cognition

EEG Methods and Analysis

Fall Semester 2026 — St. Olaf College

Location: RNS 124

Tuesday 9:35–11:00 AM — Thursday 9:30–10:50 AM
September 10 – December 15, 2026

Instructor: Dr. Brian Rivera — rivera21@stolaf.edu — Office: RNS 160A

Office Hours: Mondays and Wednesdays 1:00–2:30 PM or by appointment

Credits: 1.00 — Prerequisites: PSYCH 125 and PSYCH 230 or consent

1 Course Description

In this seminar, you will learn to process and interpret **electroencephalography (EEG)** and **event-related potential (ERP)** data using MATLAB. The course has two tracks:

- **ERP Theory:** Each week, one student leads discussion on a major ERP component: its cognitive significance, neural generators, and research applications.
- **Technical Skills:** Weekly MATLAB labs walk you through the complete EEG/ERP processing pipeline, from raw data to publication-quality figures.

We will use the **ERP CORE** dataset, which contains data from seven classic ERP paradigms. With seven students in the seminar, each of you will own one of those seven components: you will present on it, and you will analyze it for your final project. You will work with real electrophysiological recordings and learn to justify your analytical decisions at every processing step. For your final project, you will conduct a complete ERP analysis of your assigned component in a MATLAB Live Script, combining what you've learned about the theory with your new technical skills.

2 Learning Objectives

By the end of this course, you will be able to:

1. **Explain** how EEG signals arise from neural activity and what they tell us about cognitive processes.
2. **Characterize** major ERP components (N170, MMN, N2pc, N400, P3b, LRP, ERN): their latency, polarity, scalp distribution, neural generators, and cognitive significance.
3. **Implement** a complete EEG preprocessing pipeline in MATLAB: filtering, re-referencing, epoching, artifact detection, and baseline correction.
4. **Justify** your analytical decisions based on component characteristics and research questions.
5. **Create** publication-quality ERP waveform plots and topographic maps.

6. **Conduct** appropriate statistical analyses of ERP data and interpret results.
7. **Evaluate** ERP methodology in published research, identifying strengths and limitations.
8. **Communicate** electrophysiological concepts clearly in writing and presentations.

3 Required Materials

Software:

- **MATLAB:** Installed and working by Lab 1 (Sep 17). St. Olaf does not have a campus MATLAB license, so start with the free 30-day trial at <https://www.mathworks.com/campaigns/products/trials.html>, then purchase a student license (~\$99 for annual license) at: <https://www.mathworks.com/store/>
- **EEGLAB:** Free. https://eeglab.org/tutorials/01_Install/Install.html (installed together in Lab 2)
- **ERPLAB:** Free. <https://github.com/ucdavis/erplab/releases> (installed together in Lab 2)

Dataset:

- **ERP CORE:** Kappenman, E., Farrens, J., Zhang, W., Stewart, A. X., & Luck, S. J. (2021). ERP CORE: An open resource for human event-related potential research. *NeuroImage*, 225, 117465. <https://doi.org/10.1016/j.neuroimage.2020.117465>
- **Getting the data:** Labs use two subjects from my fraction comparison EEG study (BrainVision .eeg/.vhdr/.vmrk files with an event-code key), posted on Moodle in Lab 2. The ERP CORE files for your component are posted when you begin your final project; you do not need to download the full OSF archive. Keep the data on your laptop and in one backup location.

Readings: Primary literature will be selected by student presenters and approved by the instructor. Additional methodological readings will be provided on Moodle. If you are not sure how to find primary literature, ask! I am happy to help, and the reference librarians are excellent resources.

Equipment: Laptop capable of running MATLAB (minimum 8GB RAM recommended).

4 Important Dates

Fall Semester 2026:

- | | |
|------------------------------|---------------------------------|
| • First day of class: Sep 10 | • Thanksgiving Break: Nov 25–29 |
| • Last day to add: Sep 23 | • Last day of class: Dec 15 |
| • Fall Break: Oct 10–13 | • Final exams: Dec 17–22 |
| • Last day to drop: Nov 11 | |

5 Course Schedule

5.1 Session Structure

Tuesday Sessions (85 min): ERP Theory

- Student-led presentation on ERP component (40–50 min)
- Discussion of the two papers, led by presenter (25–30 min)
- Synthesis and preview of Thursday lab (10 min)

Thursday Sessions (80 min): Technical Skills

- Brief review and connection to theory (10 min)
- Hands-on MATLAB lab with guided exercises (60 min)
- Wrap-up and troubleshooting (10 min)

5.2 Weekly Schedule

Wk	Dates	Tuesday (Theory)	Thursday (Lab)	Due
Foundation (Weeks 1–4)				
1	Sep 10	—	Course Introduction & Syllabus; — Component Assignment	
2	Sep 15/17	ERP Fundamentals & The EEG Signal	Lab 1: MATLAB Introduction	Lab 1
3	Sep 22/24	ERP Methodology Overview	Lab 2: EEGLAB/ERPLAB In-stall & Data Loading	Lab 2
4	Sep 29/Oct 1	ERP Design & Literature Search	Lab 3: Filtering	Lab 3
ERP Components & Processing Pipeline (Weeks 5–12)				
5	Oct 6/8	Lab 4: Re-referencing (Tuesday)	Lab 5: Bin Assignment & Epoching	Labs 4, 5
<i>Fall Break: October 10–13 (No Class Tuesday Oct 13)</i>				
6	Oct 15	—	N400 – Semantic Processing (Thursday)	—
7	Oct 20/22	N170 – Face Perception	Lab 6: Channel Interpolation	Lab 6
8	Oct 27/29	P3b – Context Updating	Lab 7: Artifact Detection	Lab 7
9	Nov 3/5	MMN – Auditory Oddball	Lab 8: ERP Scoring & Measurement	Lab 8
10	Nov 10/12	ERN – Error Monitoring	Lab 9: Statistical Analysis	Lab 9, Proposal
11	Nov 17/19	N2pc – Visual Attention	Lab 10: Plotting & Scripting	Lab 10
12	Nov 24	LRP – Motor Preparation	—	—
<i>Thanksgiving Break: November 25–29 (No Class Thursday Nov 26)</i>				
Final Project Development (Weeks 13–15)				
13	Dec 1/3	Project Workshop: Integration	Troubleshooting & Figure Workshop	—
14	Dec 8/10	Project Work Session & Dry Runs	Final Presentations I (3 students)	Draft Due (Dec 7)
15	Dec 15	Final Presentations II (4 students) & Course Evaluation	—	—
Finals Week (Dec 17–22)				
	Finals	No exam; no class meeting	Final due on this course's registrar-assigned final date	Final + Reflection

Presenters for each component are posted on Moodle. The N400 is presented on a Thursday, the first meeting after Fall Break, so week 5 has two labs.

5.3 ERP Component Assignments

Each student will be assigned one ERP component from the ERP CORE dataset. You will present on this component during weeks 6–12 and analyze it for your final project. Additionally, each student will be assigned one **preprocessing step** to explain in detail during their final presentation (see Final Presentation requirements). Components were assigned by preference ranking in the first weeks of the semester; assignments are posted on Moodle.

Component	Paradigm	Cognitive Domain	Week	Presents	Preproc. Focus
N400	Word Pair	Semantic Processing	6	Oct 15 (Thu)	Chan. Interpolation
N170	Face Perception	Visual/Face Processing	7	Oct 20	Filtering
P3b	Visual Oddball	Context Updating	8	Oct 27	Re-referencing
MMN	Passive Oddball	Pre-attentive Processing	9	Nov 3	Epoching & Bins
ERN	Flankers	Error Monitoring	10	Nov 10	Scoring & Statistics
N2pc	Visual Search	Spatial Attention	11	Nov 17	Artifact Detection
LRP	Flankers	Motor Preparation	12	Nov 24	Baseline & Averaging

5.4 MATLAB Lab Sequence

Labs follow the EEG/ERP processing pipeline, building skills sequentially. Each lab includes a specific deliverable due by the end of class.

Lab	Topic	Skills Covered	Deliverable
1	MATLAB Introduction	Syntax, variables, arrays, functions, plotting	Script: plot sine wave with labels
2	Install & Data Loading	EEGLAB/ERPLAB setup, loading .set files	Screenshot: loaded dataset info
3	Filtering	High-pass, low-pass, filter types	Script: filter data, plot PSD before/after
4	Re-referencing	Average reference, mastoids, effects on ERPs	Script: re-reference, plot comparison
5	Bin Assignment & Epoching	Event codes, bin descriptors, epochs	Script: create epochs for 2 conditions
6	Channel Interpolation	Bad channel detection, interpolation	Script: identify & interpolate 1 channel
7	Artifact Detection	Voltage thresholds, blinks, rejection	Script: detect artifacts, report % rejected
8	ERP Scoring	Peak amplitude, mean amplitude, latency	Script: measure component, export values
9	Statistical Analysis	t-tests, ANOVA, effect sizes	Script: run t-test, report stats & effect size
10	Plotting & Scripting	Publication figures, automation	Script: create formatted ERP figure

6 Assessment and Grading

6.1 Grade Distribution

Point Breakdown (1000 points total)

Assessment	Points	%
Student-Led Presentation	200	20%
Weekly MATLAB Labs	150	15%
Peer Feedback (Given)	50	5%
Biweekly Reflections	35	3.5%
Participation	65	6.5%
Final Project	500	50%
Total	1000	100%

Final grades are not rounded. 929 points is an A-, 899 is a B+, and so on.

Where your grades are: everything is posted in the Moodle gradebook. Participation is assessed holistically at midterm and at the end of the semester; ask me at any time where you stand.

Grading Scale

Grade	Points
A	930–1000
A-	900–929
B+	870–899
B	830–869
B-	800–829
C+	770–799
C	730–769
C-	700–729
D+	670–699
D	600–669
F	Below 600

7 Assessment Details

7.1 Student-Led Presentation (200 points)

Each student presents on their assigned ERP component and leads discussion on a primary research paper.

Component	Points	Description
Paper Selection	20	Submit two papers on Moodle one week before your presentation (the previous Tuesday, 11:59 PM): one seminal and one recent (2021 or later). I approve them within 24 hours and post both for the class
Presentation	100	40–50 minute presentation covering component characterization, neural generators, cognitive function, key findings, and experimental paradigm
Discussion Questions	20	3–5 thoughtful questions to lead class discussion; posted on Moodle with the handout
Reference Handout	20	One-page summary for classmates (latency, polarity, topography, key facts); due on Moodle the night before your presentation (Monday, 11:59 PM)
Peer Feedback Received	40	Mean of the 6 peer evaluations (five criteria on a 1–5 scale, 25 raw points), scaled to 40
Total	250	

Your presentation should cover:

- **Component basics:** Latency range, polarity, scalp distribution
- **Neural generators:** Where does it come from? (cite neuroimaging or lesion studies)

- **Cognitive function:** What process does this component reflect?
- **Key findings:** 3–4 important studies using this component
- **Paradigm:** The classic task used to elicit it

Scoring: the 100 presentation points are split component characterization 25, neural generators 20, paradigm and key findings 25, discussion leadership 20, delivery 10. The full rubric, including the handout, discussion question, and peer evaluation scales, is posted on Moodle with the presentation instructions.

7.2 Weekly MATLAB Labs (150 points)

Ten hands-on labs building EEG/ERP analysis skills sequentially.

Scoring: each lab is out of 15: the script runs and produces the deliverable (8), the settings are correct for the task (4), and the code is readable (3). The lab rubric is posted on Moodle with Lab 1.

Lab submissions are uploaded to Moodle (the script or screenshot named in the lab table) by the end of each Thursday lab session unless otherwise specified. **Late labs** are graded on the same scale and capped at 8 of 15.

7.3 Peer Feedback Given (50 points)

You will evaluate each classmate's presentation.

Component	Points	Description
Complete Evaluations	42	7 points each for 6 presentations (all except your own)
Quality of Feedback	8	Thoughtful, constructive, specific comments
Total	50	

Evaluation Format: Rate each presentation on a 1–5 scale for clarity, depth of content, engagement, visual aids, and discussion facilitation. Include “Glow and Grow” comments (one strength, one suggestion). Evaluations are a Moodle form, due by 11:59 PM on the day of the presentation; the presenter sees the anonymized ratings and comments after I post them.

7.4 Biweekly Reflections (35 points)

Every other week, you'll submit a short reflection so I can see how things are going and adjust the course if needed.

Component	Points	Description
7 Reflections	5 pts each	Pass/fail based on completion and thoughtfulness
Total	35	

Schedule: Due on Moodle by Sunday 11:59 PM at the end of weeks 2, 4, 6, 8, 10, 12, and 14: Sep 20, Oct 4, Oct 18, Nov 1, Nov 15, Dec 1 (Tuesday, after Thanksgiving), and Dec 13.

Prompt (100–150 words): Reflect on the past two weeks. What concepts clicked? What's still confusing? How are you feeling about the technical skills vs. the theory? Any concerns or suggestions?

Grading: Pass/fail. Full credit for genuine, thoughtful engagement. These are confidential and help me support your learning.

7.5 Participation (65 points)

In a seminar this size, everyone's engagement matters.

Component	Points	Description
Discussion Engagement	30	Thoughtful contributions to paper discussions; demonstrates reading preparation
Peer Support	25	Helping classmates troubleshoot code; collaborative learning
Attendance	10	Two unexcused absences are free; each one after that costs 3 points
Total	65	

Discussion engagement and peer support are assessed holistically twice: a midterm score entered over Fall Break (Oct 10–13), and a final score after the last class. Ask me at any time where you stand. See the Attendance Policy below.

7.6 Final Project: ERP Component Analysis (500 points)

You will analyze your assigned ERP component from start to finish in a MATLAB Live Script, showing that you understand both the theory and the technical methods.

Project Point Breakdown

Deliverable	Points	% of Project	Due Date
Project Proposal	50	10%	Week 10 (Nov 12)
Draft Live Script	50	10%	Dec 7 (11:59 PM)
Final Presentation	100	20%	Dec 10 or Dec 15 (in class)
Final MATLAB Live Script	250	50%	Registrar's final date
Self-Reflection Essay	50	10%	Registrar's final date
Total	500	100%	

Project Proposal: Analysis Plan (50 points)

Your component is already assigned, so the proposal is an analysis plan, not a justification of the choice. In 250–400 words, state the two conditions you will contrast and the predicted difference; the electrode site(s) and the measurement window, with a citation for each; the scoring method (mean amplitude, peak amplitude, or latency) and why it suits your component; the statistical test; and the preprocessing decisions you expect to matter most for your component. Each choice gets a one-line rationale. This becomes the skeleton of sections C through E of your Live Script. I return feedback within one week.

Draft Live Script (50 points)

Due Dec 7 at 11:59 PM as an .mlx file. A draft has sections A through D complete and running on your component's data, and sections E and F at least outlined. I return written comments before your presentation day; Dec 8 is a work session for acting on them and a dry run of your preprocessing teach-through.

MATLAB Live Script Requirements (250 points)

Your Live Script should integrate code, output, and narrative text. Section requirements below; the detailed rubric is posted on Moodle with the project instructions.

Section	Points	Requirements
A. Introduction & Theory	35	Component characterization, neural generators, cognitive function, review of 3–4 seminal studies, predictions (750–1000 words; min. 5 sources)
B. Data Loading & Inspection	25	Load dataset, display properties, show channel locations, visualize raw data with justifications
C. Preprocessing Pipeline	100	Filtering, Re-referencing, Artifact Detection, Epoching, Baseline Correction, Averaging; each with rationale, implementation, parameter justification, quality checks
D. Visualization & Measurement	50	Publication-quality ERP waveforms, topographic maps, component measurements with justifications
E. Statistical Analysis	25	Appropriate tests, effect sizes, visualization, interpretation
F. Discussion	15	Results interpretation, limitations, future directions (500–750 words)
Total	250	

Final Presentation (100 points)

Format: 15 minutes + 5 minutes Q&A, in class during the last two meetings: three students present on Thursday Dec 10 and four on Tuesday Dec 15. Presenting order is posted on Moodle before Dec 10 and is fixed. You present from your draft Live Script; the final version, with any revisions prompted by the presentation, is due on the date of this course’s registrar-assigned final.

Your presentation has two parts: (1) presenting your analysis results, and (2) **teaching one preprocessing step** by walking through your code.

Slide outline:

- **Slide 1:** Title, name, ERP component
- **Slides 2–3:** Background: what is your component and why does it matter?
- **Slides 4–6: Preprocessing Deep Dive:** walk through your assigned preprocessing step line-by-line. Show your code, explain it, justify your parameters, and show quality checks.
- **Slides 7–9:** Results (ERP plot, topographic map, statistics)
- **Slide 10:** Interpretation
- **Slide 11:** Limitations and future directions

Preprocessing Step Assignments:

- N170 → Filtering
- P3b → Re-referencing
- MMN → Epoching & Bin Assignment
- N400 → Channel Interpolation
- N2pc → Artifact Detection
- LRP → Baseline Correction & Averaging
- ERN → Scoring & Statistics

Scoring: preprocessing teach-through 40, results 25, interpretation and Q&A 25, delivery 10. The full rubric is posted on Moodle with the project instructions.

Note: The presentation verifies you understand your analysis. By having each student teach a different preprocessing step, the class reviews the entire pipeline together.

Self-Reflection Essay (50 points)

Length: 400–500 words

Address: What was hardest? What surprised you? If you used AI or outside help, how did you verify it was correct? What did you learn from teaching your preprocessing step? What would you do differently next time?

8 Course Policies

8.1 Academic Integrity & Plagiarism

Plagiarism is not only taking someone else's ideas and presenting them as your own without proper citation, it is also changing a word or two and using citations inappropriately. In a methods course, the same applies to code: submitting an analysis pipeline you copied and do not understand is plagiarism.

I take academic dishonesty very seriously and any instances will result in, at minimum, a failing grade for that assignment and possibly a failing grade in the course. For complete details on St. Olaf's academic integrity policy, see: <https://wp.stolaf.edu/thebook/plagiarism/>

8.2 AI Tool Usage Policy

Student process augmented by AI usage & exploration. In this course, your work may incorporate LLM usage and exploration, with caveats explained below. Using AI, in conjunction with your own critical thinking, may present some valuable learning experiences. AI may be used extensively as a co-agent on assignments in this course to augment learning and critical thinking.

AI is still very new, and we are all discovering it (professors and students alike). We will discuss innovative or challenging uses of AI together as a community. Feel free to propose uses of these new tools to your instructor. Let me know if you have any ethical or critical concerns at any point in the semester.

You may use AI and advanced machine learning tools such as Gemini, ChatGPT, Imagen, Dall-E, Suno, or Veo with proper documentation and credit, as described in our class.

This means you are responsible for:

- Critically engaging with readings, lectures, discussions, and other materials.
- Using this learning to develop approaches, research, and insights that become the submitted work for the course.
- Composing all assignment drafts deliberately and critically, even when AI is involved.
- Revising and iterating assignments, including through feedback from peers and your professor.
- Ensuring that any AI practices adhere to the uses and responsibilities below.

You may use AI to:

- Get started by brainstorming initial questions or formulating research plans.
- Suggest concepts, strategies, or outlines to inform your drafting process.
- Generate draft or design components that you then transform, demonstrating advanced skill-building.
- Get feedback, guidance, examples, and counter-perspectives that augment your choices during revision.
- Check for grammar and style, but skeptically.
- Ask for additional explanations or tutoring.

If you use AI, you should:

- Provide documentation tracking how AI augmented your thinking and how you transformed what it generated.
- Verify and cite all information (numbers, data, quotations, sources, claims, etc.) per standards outlined in class.

- Avoid using AI on quizzes or exams.
- Think carefully about the ethics of AI and work to minimize harm.

By enrolling in this course, you pledge to uphold these policies. You recognize that failure to do so means that you have likely cheated, plagiarized, or falsified information in violation of St. Olaf's Academic Integrity policy and Honor Code.

In this course specifically: there are no quizzes or exams; the final presentation is the oral exam, and that is where the line sits. Use AI as much as you like to learn, debug, and explore, but you need to understand your own work.

Permitted:

- Using AI tools (ChatGPT, Claude, Gemini, MATLAB Copilot, etc.) to write, debug, explain, and refactor code
- Using AI as a tutor for ERP theory, MATLAB, and statistics, and to generate practice questions before your presentation
- Generating a first draft of a figure, a script section, or an outline that you then transform and verify
- Consulting online resources, documentation, and tutorials; discussing approaches with classmates

Required:

- **You must understand every line of code in your script**, whoever or whatever wrote it
- You must be able to explain and defend your analytical decisions in the presentation
- You must cite any substantial code borrowed from external sources
- You must document how AI augmented your work and how you transformed what it generated (a short AI-use note at the end of your Live Script, and in your self-reflection essay)
- You must verify every claim, number, and citation AI gives you; a hallucinated reference in your theory section is a plagiarism case, not a typo

Prohibited:

- Submitting code you cannot explain
- Submitting AI-generated theoretical text, discussion questions, or handouts you have not verified and made your own
- Misrepresenting the extent of assistance received

The final presentation serves as an oral exam to verify your understanding. If you cannot explain your code or justify your decisions during the presentation, your project grade will be significantly reduced regardless of Live Script quality.

Instructor's AI Use: I use AI extensively for course planning, research, coding, and formatting. I've built a library of digital diagrams using AI, including an interactive page for each step of the ERP pipeline, that I will be sharing throughout the semester. I am responsible for verifying all AI-generated content for accuracy before sharing.

Important Notes: This policy applies to this course only. Other professors may have different policies, and we will adapt to any college-wide policy changes.

8.3 Attendance Policy

Attendance is essential. Each session builds on the last, and with seven students, your presence matters. I expect everyone to be present and ready to participate every day. Attendance is 10 points of your participation grade: two unexcused absences are free, and each one after that costs 3 points.

- **Excused absences:** Whether an absence is excused is my call. I do not require a doctor's note, and I will take you at your word. That trust is the whole system: misrepresenting the reason for an absence, a late assignment, or a missed presentation is academic dishonesty and will be handled under the Academic Integrity policy above, which means at minimum a failing grade for the work involved.
- **Presentation days:** Missing your assigned presentation day requires rescheduling and may result in a point deduction. Since each week's lab follows that week's component, a missed presentation affects the whole class; if you know you cannot make your day, trade with a classmate and email me before your paper selection is due (one week before).
- **Lab days:** Missed labs must be completed independently; office hours are available for support.
- **Tardiness:** Please arrive on time. It is disruptive to arrive late, and on Tuesdays it is unfair to the presenter.
- **Communication:** Let me know about anticipated absences sooner rather than later. Life happens.

8.4 Late Work & Make-up Policy

- **MATLAB Labs:** Late submissions are graded on the same scale and capped at 8 of 15
- **Presentation materials:** Paper selection (one week before), handout and discussion questions (the night before) must be on time so the class can prepare; late submissions lose those points
- **Draft Live Script:** Due Dec 7 at 11:59 PM so I can read it before the presentations; a late draft gets partial credit and no written feedback before you present
- **Reflections:** Not accepted late; each is open for the full two weeks, so plan for it
- **Final Project:** Late submissions require prior arrangements. Contact me before the deadline if you anticipate problems; an extension is at my discretion and the honesty rule in the Attendance Policy applies
- **Final Presentation:** No make-up. Missing your presentation slot is 0 for the presentation, and the Live Script cannot receive full credit without the oral defense
- **Communication is key:** If you know you'll be unable to complete work on time, talk to me about it ahead of time!

8.5 Technology Requirements & Classroom Expectations

- **Laptop:** Required for all Thursday lab sessions; minimum 8GB RAM recommended for MATLAB
- **Software:** MATLAB installed and running by Lab 1 (Sep 17); EEGLAB and ERPLAB are installed together in Lab 2 (Sep 24)
- **Backup:** Maintain backups of all code and data; technical issues are not grounds for extensions
- **Phones:** Silenced and put away during class. On lab days your laptop is for MATLAB; on presentation days it stays closed unless the presenter asks otherwise
- **Respect:** Only one person speaks at a time. Respect the ideas and contributions of your classmates; on Tuesdays the presenter is the class expert, and the discussion is theirs to lead

Academic Commitment: Students are expected to devote a minimum of 3 hours outside class per hour in class. In this course that time goes to preparing for discussion, finishing labs, and building your Live Script.

My Responsibilities:

- Be reasonably available to answer questions and help you formulate questions.
- Help you explore your interests in this fascinating material.
- Grade assignments fairly and in a timely fashion.
- Respect your ideas and contributions.

9 Support & Resources

9.1 Technical Support

- **Weekly lab hours:** Thursday sessions include troubleshooting time
- **Office hours:** Mondays and Wednesdays 1:00–2:30 PM in RNS 160A, or by appointment
- **Online resources:** EEGLAB tutorials, ERPLAB documentation, the ERP pipeline diagram pages, and the course Moodle forum

9.2 Disability Accommodations

If you have a documented disability for which accommodations may be required in this class, please contact the Academic Support Center as soon as possible to discuss accommodations. If you have already arranged accommodations through Student Accessibility Services, please arrange for submission of your accommodation letter within the first two weeks of class.

Accommodations will only be provided after the letter is submitted with sufficient lead-time for me to arrange them. Although I will receive the letter electronically, I expect you to initiate a conversation with me about the accommodations. Please do this by the end of the second week so that we can plan for presentations.

Contact: Academic Support Center, Buntrock 108, 507-786-3288

9.3 Additional Support & Resources

Getting Help: Should you encounter difficulties with course material, please contact me during office hours, via email, or after class. I am committed to supporting your academic success.

Communication: Course announcements go out on Moodle. I answer email within one business day; I do not answer on weekends, so a question about a Monday deadline should reach me by Friday.

- Use subject line: “PSYCH 390: [Your Topic]”
- For code debugging, include error messages and screenshots
- For quick questions, see me after class or in office hours

Study Resources:

- Visit the Writing Desk (Tomson 157) for help with the theoretical sections of your Live Script and your presentation: <https://wp.stolaf.edu/academic-support/writing-help/>

- Research librarians can help you find and cite primary literature for your presentation and project: <https://www.stolaf.edu/library/research/appointments/>
- Help each other troubleshoot; that is part of your participation grade

9.4 College Statements

Inclusivity: In keeping with St. Olaf College’s mission statement, this class strives to be an inclusive learning community, respecting those of differing backgrounds and beliefs. As a community, we aim to be respectful to all citizens in this class, regardless of race, ethnicity, nationality, religion, gender, sexual orientation, or disability.

Mental health: If you are experiencing mental health symptoms or issues, multiple resources are available to you on campus, including the Dean of Students office (<https://wp.stolaf.edu/deanofstudents/>), the Counseling Center (<https://wp.stolaf.edu/counseling-center/>), and TimelyCare (<https://wp.stolaf.edu/timelycare/>). Please do not hesitate to come see me if you need assistance connecting with the support you need.

Required referrals: You are welcome to speak to me about any issues that arise for you in the context of this course, including any circumstances outside of class that may be affecting your classroom experience or academic performance. However, please be aware that I am required to refer to the college’s Director of Equal Opportunity any information students share about experiences of sexual harassment, sexual assault, or dating violence. I am also expected to share any disclosures of incidents of bias, harassment, or discrimination to the Bias Response Team. Confidential resources that do not have this obligation include the Counseling Center, College Ministry (<https://wp.stolaf.edu/ministry/>), the Sexual Assault Resource Network (<https://pages.stolaf.edu/sarn/>), Health Services (<https://wp.stolaf.edu/health/>), and TimelyCare. More information: <https://wp.stolaf.edu/title-ix/support/> and <https://wp.stolaf.edu/community-belonging/>.

9.5 Final Presentations & Finals Week

There is no final exam and no meeting during finals week. Final presentations take place in class on Dec 10 and Dec 15, and both days are required for everyone: on the day you are not presenting you are the audience, and hearing your classmates teach their preprocessing steps is how the class reviews the whole pipeline together. Missing your presenting day is a 0 for the presentation; there is no make-up. Your final Live Script (the .mlx file plus a PDF export) and self-reflection essay are due on Moodle by 11:59 PM on the date the registrar assigns as this course’s final (posted on Moodle once the finals schedule is out).

9.6 Course Philosophy

By the end of the semester, you will understand not only what ERP components measure, but *how* to measure them and *why* each analytical decision matters. This course works best when everyone is engaged. With only seven students, we rely on each other; you will each become the class expert on your component. The labs build on each other, so staying current is important. The skills you develop here (programming, data analysis, methodological reasoning, scientific communication) will serve you in any research or data-intensive career.

Syllabus subject to change with advance notice. Updates announced in class and on Moodle. All dates and topics are subject to change; we might need to slow down or explore a different topic. Presentation dates will only be changed by agreement of the entire class.

Updated 9/9/2026